

Practice Quiz: Systems (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. A dynamical system is a system whose: A) Structure never changes B) State evolves over time according to rules C) Components are always neurons D) Behavior is completely random
2. Neural oscillations are: A) Random electrical noise in the brain B) Rhythmic patterns of neural firing produced by interactions between excitatory and inhibitory neurons C) Only measurable with expensive equipment D) The same across all brain states
3. Gamma oscillations (~40 Hz) are associated with: A) Deep sleep B) Attention, binding of features, and conscious awareness C) Relaxation D) Motor planning exclusively
4. An attractor is: A) A magnet inside the brain B) A stable pattern of activity that a dynamical system tends to settle into C) A type of neuron D) Something that only exists in mathematical models
5. Memory recall as an attractor means: A) The brain physically moves toward a memory B) Neural activity is drawn toward a stored pattern of activation C) Memories are stored in single neurons D) Recall is always accurate
6. A seizure can be understood as: A) Normal brain function B) A breakdown of excitation-inhibition balance leading to runaway synchronization C) Too little neural activity D) A fixed-point attractor
7. The Markov Blanket of a neuron consists of: A) Its cell membrane only B) Its dendrites (sensory inputs), axon terminals (active outputs), and internal states (membrane potential) C) All the neurons it has ever communicated with D) Only its DNA

Part B: Short Answer

- 1.** Explain how excitatory and inhibitory neurons work together to produce oscillations. What happens if only one type is present?
- 2.** Choose one neural oscillation frequency band (alpha, beta, gamma, or theta) and explain what cognitive function it supports, when it is most prominent, and what happens when it is disrupted.
- 3.** Explain the attractor metaphor using a physical example (e.g., ball in a valley) and then apply it to one specific neural phenomenon (e.g., memory recall, resting state, or habitual behavior).